

Case report

The impact of climate change on psychiatric decompensation: A case report using the climate biopsychosocial framework

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ABSTRACT

Introduction: Anthropogenic fossil fuel emissions are warming the planet quickly, leading to climatic instability and more extreme weather events that impact patient health.**Case Reports:** This case study examines a 44-year-old Asian-American male patient with bipolar disorder who decompensated psychiatrically associated with the impacts of two sequential climate-change fueled natural disasters.**Discussion:** The classic biopsychosocial (BPS) model was developed during our previous era of climatic stability and is limited in its ability to describe the variables that caused the patient in this case report to decompensate. We introduce a Climate Biopsychosocial (CBPS) model to account for climate change's enormous impact on patient health.**Conclusion:** The CBPS model is a framework that expands patient care and research pathways.

1. Introduction

George Engel introduced the biopsychosocial (BPS) model in 1977 [1] after recognizing that the biomedical model of disease was too narrow to account for the constellation of variables that impact health. At that time, carbon dioxide (CO₂) emissions' global warming effect was understood among scientists, but the impacts were not yet being widely felt [2]. Dr. Engel's generation of physicians was working at the end of a 10,000 year period of climate stability which supported our modern civilization (Fig. 1).

In the 1980s, scientists began detecting a discernable warming trend in climate data, as can be seen in Fig. 2 [3]. By 1995, the International Panel on Climate Change (IPCC) concluded that fossil fuel emissions were warming the climate [4]. Today, research demonstrates how greenhouse gas emissions have destabilized our climate, and contributed to, for example, a 377 % increase in the frequency of compound drought-heatwaves from 1981 to 2020 [5] and a 60 % increase in extreme precipitation events between 1984–2013 [6]. Event attribution studies can determine our emissions contribution to specific event probabilities, such as a 35 % increased likelihood of the 2025 Los Angeles, California Palisades Fire [7].

A growing body of literature illustrates climate change's acute and

chronic impact on mental health. Acute risks include a 3.0 odds ratio of death for patients with schizophrenia during heat waves [8], a relative risk of 1.073 for hospitalizations for schizophrenia patients after severe flooding events [9], a 0.7 % increase in suicide for a 1°C increase in monthly average temperatures [10] and 12.8 %–26 % increases in post-traumatic stress disorder (PTSD) 6 months after direct exposure to life-threatening wildfires [11]. Chronic risks include function-limiting climate anxiety among children and young adults as well as wildfire PM_{2.5} air pollutants increasing dementia risk with a hazard ratio (HR) of 1.05 [12,13].

The BPS model historically has accommodated individual disaster events - a single wildfire or hurricane - as an isolated psychosocial stressor. However, climate change, because of the breadth and depth of its impacts, exceeds the model's explanatory effectiveness. For example, sea level rise is an influential factor chronically across BPS domains: biologically, encroaching salinity damages water supplies and nutritional quality; socially, communities and livelihoods are disrupted; psychologically, distress at scale overwhelms individual and community coping mechanisms [14]. Acutely, increasingly frequent, severe, and compound disasters induce non-linear health impacts. For example, compound fire and drought events lead to a threefold increase in psychiatric ED visits compared to non-compound events [15]. The BPS

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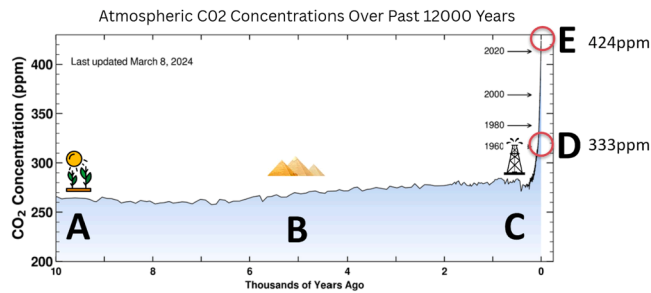


Fig. 1. Atmospheric CO₂ Concentrations Over Past 12,000 Years. A is the birth of agriculture. B is the construction of the great Egyptian pyramids. C is the first commercial oil well in Pennsylvania in 1859. D is the publication of the biopsychosocial model, when atmospheric CO₂ concentrations were 333 parts per million (ppm). E is today, when CO₂ concentrations are 424 ppm. Graph and CO₂ data courtesy of <https://keelingcurve.ucsd.edu/> (accessed 3/10/24).

model, while foundational, requires a new dynamic causal climate variable to remain clinically and conceptually adequate today.

We demonstrate this via a case report, in which we describe a patient who decompensated psychiatrically related to the indirect impacts of two sequential climate events. The first, the August 8th, 2023 Lahaina fire, resulted from climate change-exacerbated drought, powerful winds, and flammable invasive grass [16,17] in Hawaii, which has a 375 % increased regional long term fire risk due to anthropogenic emissions [18]. The second, the August 20, 2023 tropical storm Hilary, fueled by a combination of anthropogenic atmospheric and sea surface temperature warming, and a strong El Niño, obliterated rainfall records and brought with it Southern California's first tropical storm warning [19,20].

2. Case presentation

On August 21, 2023, a 44-year-old Asian-American male with a medical history significant for bipolar disorder, unspecified anxiety, stroke two years prior, psychogenic tremor, multiple past suicide attempts by overdose, four previous psychiatric hospitalizations, and regular nicotine and cannabis use was brought to the emergency

department by ambulance after disclosing to his father that he intended to overdose and cut himself. Upon evaluation in the ED, the patient endorsed severe anxiety and suicidal ideation that had been increasing over the previous two weeks and worsened acutely 24 hours prior to presentation. Denying any triggers, the patient did not know why his symptoms had worsened.

The patient had been taking his long-term psychiatric medications, including lurasidone, lamotrigine, escitalopram, gabapentin, propranolol and buspirone daily as prescribed. Psychiatrist and therapy notes from earlier in August indicated the patient was at his baseline with stable sleep and well controlled anxiety.

The patient is divorced, lives with his father, is unemployed, and his daughter, who lives across the country, was psychiatrically hospitalized and subsequently discharged for suicidal ideation one month prior to the patient's admission. Additionally, the patient reported that his mother had passed away several years prior and he missed her greatly.

The patient's father reported that until two weeks ago the patient had been doing well, and they had been preparing for a move to Lahaina, Maui, to be closer to relatives. The catastrophic Lahaina fire on August 8th and 9th completely upended their plans, and they took their house off of the market. With this drastic life plan change, the father reported an increase in the patient's anxiety. The father additionally reported that the patient's anxiety worsened further on the day prior to admission, after the patient learned that his brother had been impacted by tropical storm Hilary.

3. Discussion

This patient experienced three significant stressors in August 2023. The first was his daughter's psychiatric hospitalization for suicidal ideation. Notably, both the patient's psychiatrist and therapist commented in their notes that despite this stressor the patient maintained a stable mood, good sleep, and manageable anxiety. The second and third stressors were the Lahaina fire and tropical storm Hilary. The patient's father believed that the confluence of disaster impacts acutely destabilized the patient and we therefore suggest a "two hit" climate change hypothesis for this patient's decompensation.

The patient developed increasing anxiety and intermittent suicidal

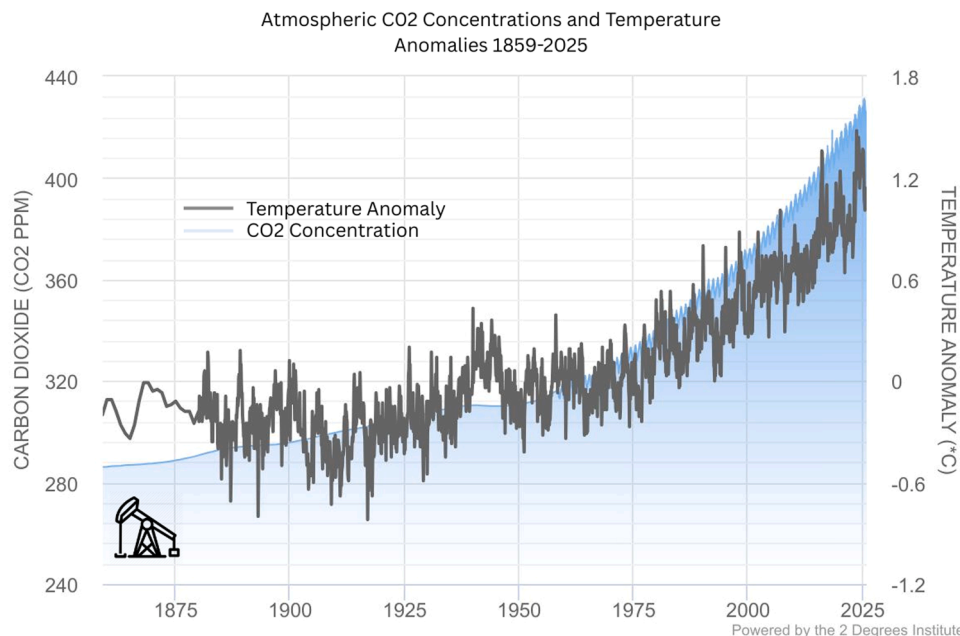


Fig. 2. Atmospheric CO₂ concentrations and global temperature anomalies 1859–2025. Atmospheric carbon dioxide (blue line, left axis) and global mean surface temperature anomaly (gray line, right axis, relative to the IPCC 1850–1900 pre-industrial baseline) are shown from the 1859 first commercial oil well in the USA. Both CO₂ and temperature rise abruptly after ~1850. Data courtesy of 2° Institute.

ideation after the family's move to Maui was derailed by the devastating Lahaina fire on August 8th, demonstrating the indirect impacts of climate change. Twelve days later, tropical storm Hilary hit Southern California where the patient's brother lived. Concerned for his brother's well-being, the patient again was indirectly impacted by climate change. The next day, August 21st, the patient presented to the emergency department. We believe that these climate change events provoked the emergency department presentation.

Under the BPS model, the patient in this case has biological risk factors including bipolar disorder, unhealthy substance use, and a history of stroke and tremor. Environmental variables like photoperiod and heat can decompensate via direct biological pathways yet seem unlikely to have contributed in this presentation, as the peak photoperiod occurred two months prior and August temperatures were within the normal range [21,22]. Psychologically, the patient demonstrates limited insight and coping mechanisms, continues to grieve the loss of his mother, struggles with his daughter's recent hospitalization, and has unemployment aggravated low self-esteem. Socially, the patient has a limited social circle, lives with his father, is divorced, and lives far from his daughter.

Under this standard formulation, we cannot understand why the patient presented to the hospital. We therefore demonstrate the clinical utility of a climate change era framework for both understanding a patient presentation and creating an appropriate treatment plan.

3.1. The climate biopsychosocial model

Over the past half century, conceptions of “environment” in health have evolved. Early psychosocial frameworks highlighted the role of relationships, culture, and community stressors in illness [23,24]. The same year that Engel introduced the BPS model, rooted in these previous

frameworks, Bronfenbrenner's *Ecological Systems Theory* situated individuals within nested social and physical environments [25]. In 1993, with increased environmental awareness partially attributable to the ozone hole, Roszak in *Ecopsychology* emphasized that human and planetary wellness co-exist on a continuum [26]. The 2005 Millenium Ecosystem Assessment introduced the “ecosystem services” concept, which was integrated into the United Nations Sustainable Development Goals in 2012 [27]. This concept contributed to the planetary-health framework which acknowledges the human well-being dependence on flourishing natural systems [28]. Dr. Moffic proposed a bio-psych-eco-social-environmental model for psychiatry, a call to consider planetary health as part of mental health [29]. Crandon developed a theoretical model of climate anxiety and coping [30] and Wang introduced a climate component within the traditional 4P case formulation via a theoretical case vignette to encourage climate change assessment in patient care [31].

The Climate Biopsychosocial (CBPS) model builds on this lineage by extending the concept of an encompassing climate context into a structured, visual framework. The key to this model can be seen in Fig. 3, which depicts the traditional biopsychosocial domains nested within the planetary climate, symbolizing that human health now is ensconced in an unstable environment. By situating these relationships within a single systems map, the CBPS model helps clinicians identify multiple level climate process-patient vulnerability interactions. Its value is conceptual and pedagogical, offering a shared language for psychiatry, public health, and climate science, and invites the field to think and communicate in a way that reflects the realities of practicing medicine in the Anthropocene.

The traditional BPS formulation identifies this patient's vulnerabilities - bipolar disorder, anxiety, substance use, social isolation and low self esteem - but not the crisis trigger. Incorporating the climate context,

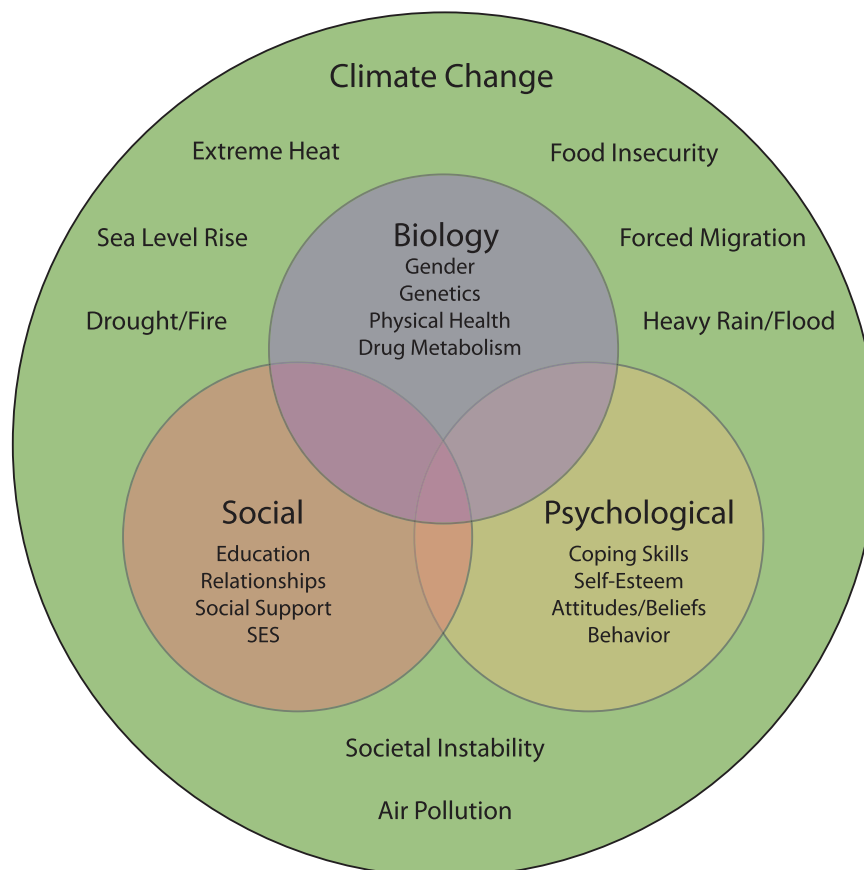


Fig. 3. The Climate Biopsychosocial Model (CBPS). The CBPS nests the biological, psychological and social components of Engel's original model within the planetary climate, symbolizing that all variables that contribute to human health are impacted by our changing climate.

the CBPS explains why the patient's symptoms escalated into a crisis on August 21st, 2023. Sequential, climate-amplified disasters functioned as compound stressors that exceeded his adaptive capacity, precipitating his emergency department presentation. The CBPS model thus reveals how environmental instability is a proximate catalyst for psychiatric decompensation, connecting individual distress to larger planetary processes.

Beyond this individual case, the CBPS model helps clinicians trace the causal chains linking climate-related stressors to health outcomes. A single step chain might involve a patient who develops anxiety and flashbacks after surviving a climate exacerbated hurricane that destroyed her home. A more complex chain could connect increased wildfire activity to insurance-market collapse [32,33] and forced migration; for example, wildfire losses destabilize insurers, leaving a patient unable to insure his home, prompting relocation, isolation, and depression. Such mapping helps clinicians and researchers see how planetary change cascades through biological, social and psychological systems, producing mental-health consequences that traditional formulation overlooks.

4. Clinical and research applications

There are manifold benefits from utilizing the CBPS model. First, the model promotes more comprehensive care by situating patients within the planetary change context. Consider a patient distressed after her home floods for the third time in Texas due to intensifying precipitation. Rather than recommending cognitive behavioral therapy and a medication adjustment, it may be more appropriate to educate her about the climate processes driving such events and to discuss adaptive planning, including the patient's perspective on risks and benefits of longer term potential relocation. Without the CBPS, we might not consider such treatment options.

Second, the CBPS model suggests new diagnostic and therapeutic frontiers. Up to one billion people may become climate refugees by 2050 [34]. Some will experience existing illness exacerbation and others will develop new adjustment or trauma symptoms [35]. If adjustment or acute stress disorder symptoms result from climate sequela, future research may support new diagnostic subtypes and treatment modalities. For example, should HEPA air purifiers be considered insurance-reimbursable medical equipment for cognitively impaired patients exposed to recurrent wildfire smoke?

Third, the model opens new research avenues. For example, both cigarette and wildfire smoke induce the cytochrome P450 (CYP) enzymes [36]. In an era of increasing wildfires, what role does prolonged wildfire smoke exposure have on inducing CYP1A2, leading to sub-therapeutic clozapine levels and psychiatric decompensation? Research could determine whether dynamic clozapine dosing adjustments during wildfires seasons improves outcomes.

5. Limitations

This report is limited by its reliance on patient self-report and parental collateral, which may be incomplete or affected by recall bias. As a single case, the temporal association between climate-amplified events and symptom escalation cannot establish causality, and the patient's multiple pre-existing vulnerabilities complicate interpretation. The CBPS model is conceptual, and its clinical utility will require evaluation in larger, more diverse samples.

6. Conclusion

Clinicians can understand many patient presentations comprehensively within the traditional BPS framework. As climate change increasingly shapes human life conditions, we require new models that account explicitly for its influence. The CBPS framework will help our patients and care systems to navigate emerging Anthropocene health

challenges.

CRedit authorship contribution statement

Jacob Yu: Writing – review & editing, Validation, Formal analysis. **Mary Sage:** Writing – review & editing. **Daniel Bernstein:** Writing – review & editing, Writing – original draft, Supervision, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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